

Prescribed Finite Étale Algebras as Full Fibers of Keller Maps with Symmetric Monodromy

Roy van Rijn
Independent researcher

26 July 2026

Abstract

Let K be a field of characteristic zero. We first prove a prescribed finite-étale algebra realization theorem: every finite étale K -algebra of rank at least three occurs as the full fiber of an explicit polynomial map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ with Jacobian one. The construction preserves the algebra itself, not only its rank or its geometric points. As a corollary, the result gives an exact fixed-degree classification: for every $N \geq 3$, the full fibers of geometric-degree- N Keller maps in $\mathbb{A}_K^3$ are precisely the rank- N finite étale K -schemes. Rank one is realized by the identity, whereas rank two is excluded by the known degree-two theorem. Thus the attainable nonzero ranks are exactly $1, 3, 4, 5, \dots$

The scheme-theoretic engine is a general localized quadratic-gauge fiber theorem. For an arbitrary target with inverse polynomial E , it identifies the fiber as

$$\mathrm{Spec} \left((K[S]/(E))[1/\overline{E}] \right).$$

Its finite-étale, or squarefree, corollary says that when E is squarefree of degree N , the localization disappears and the quotient is a full fiber of rank N . To prescribe an algebra, start with a squarefree polynomial P of degree $N \geq 3$, translate the variable so that $P'(a)P'''(a) \neq 0$, and feed $G(S) = P(a + S) - P(a)$ into the gauge. The resulting map has coordinate degree at most $6N + 2$; at a distinguished target $E = P(a + S)$, and the full fiber is naturally isomorphic to $\mathrm{Spec} K[T]/(P)$. The cubic seed $G(S) = S + S^3$ gives the recently announced three-dimensional map coordinate for coordinate.

We compute the nonproperness set of every quadratic-gauge map exactly. It is the hypersurface

$$S_{F_G} = V(\mathrm{Disc}_S(E_{\Pi, B, C})).$$

Every repeated-root block records exactly the sheets lost from the affine fiber. The slice $\Pi = 0$ has a separate Newton-polygon stratification: its generic fiber has three affine sheets and $N - 3$ sheets at infinity. In degree three the raw resultant has an extraneous factor Π , whereas in degree at least four the plane $\Pi = 0$ is a genuine nonproperness component. Explicit Laurent arcs and their valuations represent every missing branch. Since the distinguished inverse polynomial is the squarefree polynomial $P(a + S)$, the prescribed fiber lies structurally in the maximal finite-étale locus.

For every admissible seed, not merely for a generic one, the splitting field of the generic inverse polynomial over $K(\Pi, B, C)$ has geometric and arithmetic Galois group S_N . Thus every prescribed algebra occurs as a special fiber of a maximally symmetric generic cover. Over a Hilbertian field, each constructed map also has infinitely many connected full fibers whose splitting fields have group S_N .

Connectedness, field decomposition, Galois action, real signature, local points, and scalar extension therefore pass to the Keller fiber unchanged. Over $\mathbb{Q}$ this gives an explicit rank-five

fiber with points over $\mathbb{R}$ and every $\mathbb{Q}_p$ but no rational point; rank five is minimal, and one fixed Keller map contains infinitely many such fibers.

The constructive part of the argument is formalized in Lean: the automatic choice of a , the actual three-variable polynomial map, its Jacobian-one identity, the $6N + 2$ degree bound, and the natural equivalence between the literal map fiber and $K[T]/(P)$. Independent SymPy and Singular calculations check the same identities from separate implementations.

1 Introduction

A polynomial map with nonzero constant Jacobian is étale. Its fibers are therefore finite étale schemes, possibly empty. This raises a simple reverse question: given a finite étale algebra A/K , can we build a Keller map with $\text{Spec } A$ as one of its fibers?

There is one important word in that question. The fiber should be *full*: its length must equal the geometric degree of the map. Otherwise some generic inverse sheets may have escaped to infinity, and the special fiber records only part of the inverse problem. Our first and main result answers this prescribed-algebra question: every finite étale K -algebra of rank at least three occurs as a full fiber of an explicit Keller map in three variables. Together with the elementary fact that every Keller fiber is finite étale, this gives an exact fixed-degree classification: for $N \geq 3$, the full fibers of geometric-degree- N Keller maps in $\mathbb{A}_K^3$ are precisely the rank- N finite étale K -schemes. The degree-two case is empty by the known degree-two exclusion, while the identity realizes rank one.

For rank $N \geq 3$, the construction starts from a squarefree presentation

$$A \simeq K[T]/(P).$$

Choose $a \in K$ with $P'(a)P'''(a) \neq 0$ and set

$$G(S) = P(a + S) - P(a).$$

The polynomial G is inserted into one explicit three-variable quadratic-gauge map. A fixed linear change in the target makes its Jacobian one. The resulting coordinate degrees are bounded by $6N + 2$.

The family begins with the announced counterexample itself. The seed

$$G(S) = S + S^3$$

gives Alpöge's announced map coordinate for coordinate, with no conjugation or output rescaling. The higher-degree construction is therefore an extension of the announced map rather than a separate template that happens to share its Jacobian.

The displayed map is fairly large, but its inverse problem is deliberately small. It is organized around the marked coordinate

$$S = \frac{x}{1 + xy},$$

and reduces to a single polynomial equation

$$E_{\Pi, B, C}(S) = 0.$$

The general localized quadratic-gauge fiber theorem comes first. At an arbitrary target with nonzero first coordinate, it identifies the whole fiber as

$$\text{Spec} \left((K[S]/(E))[1/\overline{E'}] \right).$$

This is the natural answer even when E has repeated factors: the source map is étale, so only roots at which E' is invertible can occur. At the target used for realization, the finite-étale (squarefree) corollary applies: $E = P(a+S)$ is squarefree, hence E' is already a unit modulo E and the localization disappears.

This local description extends to an exact global boundary theorem. The Jelonek set of the quadratic-gauge map is precisely

$$V(\text{Disc}_S(E_{\Pi,B,C})).$$

Over $\Pi \neq 0$, a root block of multiplicity m accounts for exactly m missing affine sheets. On $\Pi = 0$, two roots return to the divisor $q = 0$ and one root returns to $t = 0$, while the remaining $N - 3$ roots escape. This explains intrinsically why the prescribed squarefree fiber is full: its target lies outside the nonproperness hypersurface. It also detects a necessary degree-three caveat—the raw resultant adds an extraneous factor Π even though a general target on that plane still has all three sheets.

The generic inverse problem is as large as possible in a second sense. For every admissible degree- N seed, not just a generic choice of its coefficients, the geometric and arithmetic Galois groups of $E_{\Pi,B,C}$ over the target function field are both S_N . The proof restricts to a line with fixed Π and B : a birational parametrization of the discriminant shows that the resulting polynomial is generically Morse, so its simple finite branch transpositions generate S_N . Consequently the algebra prescribed at the distinguished target lies on a cover with maximal generic monodromy. Over a Hilbertian field, the same fixed map has infinitely many connected full fibers with splitting-field group S_N .

The reconstruction is stronger than a count of geometric points. Its formulas are natural in the test algebra, so they identify the entire fiber functor, including nilpotents and every scalar extension. After translating back from S to T , the literal fiber is the original algebra $K[T]/(P)$. Nothing arithmetic has been replaced by a numerical proxy.

The proof splits into six steps. First, a coefficientwise calculation builds the all-degree map and proves its constant Jacobian and degree bound. Second, the root-to-source formulas prove the general localized quadratic-gauge fiber theorem; its finite-étale (squarefree) corollary identifies $K[S]/(E)$ with the whole scheme fiber. Third, the exact discriminant and Newton polygons compute the nonproperness hypersurface, every sheet loss, and the valuations of the missing branches. Fourth, the discriminant parametrization and the Morse-polynomial theorem give full symmetric generic monodromy. Fifth, translation turns the inverse polynomial into the prescribed P and proves the prescribed-algebra realization theorem. Finally, monogenicity handles arbitrary finite étale algebras. Separately, the Campbell–Razar–Wright theorem rules out geometric degree two and yields the attainable-rank corollary.

Once the prescribed-algebra realization theorem is proved, the arithmetic applications are immediate. An irreducible polynomial gives a connected fiber; a product gives the same field decomposition; local points, real signature, and Galois action are inherited from the original algebra. In particular, the classical Berend–Bilu quintic produces the smallest possible Hasse-principle failure, and a second construction places infinitely many such failures inside one fixed Keller map.

The constructive part is formalized in Lean. Starting from squarefree P of degree at least three, Lean chooses an admissible translation, constructs the actual $\text{MvPolynomial}(\text{Fin } 3, K)$ map and target, proves the Jacobian-one identity and the $6N + 2$ bound, and identifies the literal fiber naturally with maps from $K[T]/(P)$ into every commutative test algebra, proves that the representing quotient is finite étale, and verifies the exact specialization $G(S) = S + S^3$. The formalization promotes Π and B to independent parameters and proves that the full inverse equation is irreducible of degree N over the iterated presentation $K(\Pi, B)(C)$; its root quotient also has $\text{finrank } N$. It further proves algebraic independence of the three displayed coordinates and constructs the induced injective

pullback on rational function fields, and proves that the supplied-translation map and target commute coefficientwise with extension of the ground field, together with the tensor-product base change of the representing quotient. The remaining inputs—the explicit identification of that pullback-field extension with the formal inverse-root extension and hence with $\text{gdeg}(F_G)$, exact nonproperness and boundary accounting, monogenicity, the generic-monodromy theorem, the degree-two theorem, and the nonarchimedean number-theoretic applications—are kept separate and proved in the paper rather than presented as formalized results. For the explicit quintic, Lean additionally proves that its representing quotient is finite étale, that the literal fiber has no rational point, and that it has both a real point and a three-adic point.

2 Which finite étale algebras occur?

Let K be a field of characteristic zero. A polynomial map

$$F : \mathbb{A}_K^m \longrightarrow \mathbb{A}_K^m$$

is a *Keller map* if $\det DF \in K^*$. Its geometric degree is

$$\text{gdeg}(F) = [K(x_1, \dots, x_m) : K(F_1, \dots, F_m)].$$

A Keller map is étale and hence quasi-finite. Every fiber over a K -point is therefore a finite étale K -scheme, possibly empty.

Definition 2.1 (Keller fiber). *A nonzero finite étale K -algebra A is a Keller fiber if there are a Keller map $F : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^m$ and a point $y \in \mathbb{A}^m(K)$ such that*

$$F^{-1}(y) \simeq \text{Spec } A, \quad \dim_K A = \text{gdeg}(F).$$

The equality of the two degrees is the fullness condition. It says that the special fiber contains all generic inverse sheets. Without it, a fiber may be shorter because some sheets have escaped through the nonproperness boundary.

The construction used in the proof can be read from left to right:

$$\boxed{\text{squarefree } P(T) \longmapsto G(S) = P(a + S) - P(a) \longmapsto \tilde{F}_G^{-1}(y_{P,a}) \simeq \text{Spec } K[T]/(P).}$$

The last arrow is an isomorphism of schemes. It is not merely a matching of K -points or a count after passing to an algebraic closure.

Lemma 2.2 (The degree-two exclusion over K). *Let K be any field of characteristic zero. A Keller map $F : \mathbb{A}_K^m \rightarrow \mathbb{A}_K^m$ of geometric degree two is a polynomial automorphism.*

Proof. Put $L = K(x_1, \dots, x_m)$ and $M = K(F_1, \dots, F_m)$. The extension L/M has degree two, so it is separable and Galois. The Galois case of the Jacobian theorem was proved over $\mathbb{C}$ by Campbell and later treated algebraically by Razar and independently by Wright [3, 9, 12].

Those results are usually quoted over $\mathbb{C}$, so we record the descent to our ground field. Choose a finitely generated subfield $K_0 \subset K$ containing the coefficients of F , and view F as a map over K_0 . After shrinking the target it is finite locally free, with rank equal to its generic degree. Flat base change preserves this rank. Since the map over K has degree two, the map over K_0 and every further scalar extension also have degree two.

Choose an embedding $K_0 \hookrightarrow \mathbb{C}$. The complexified function-field extension is quadratic and Galois, so Campbell's theorem makes $F_{\mathbb{C}}$ a polynomial automorphism. Equivalently, the homomorphism

$$\varphi : K_0[Y_1, \dots, Y_m] \longrightarrow K_0[X_1, \dots, X_m], \quad Y_i \longmapsto F_i,$$

becomes an isomorphism after tensoring with $\mathbb{C}$. Faithful flatness then forces both $\ker \varphi$ and $\operatorname{coker} \varphi$ to vanish already over K_0 . Thus φ is an isomorphism, and base change gives a polynomial inverse over K . $\square$

Theorem 2.3 (Prescribed finite-étale realization). *Let $P \in K[T]$ be squarefree of degree $N \geq 3$. Choose $a \in K$ outside the finite zero set of $P'P'''$, and put*

$$G(S) = P(a + S) - P(a).$$

Let $F_G = (\Pi, B, C)$ be the quadratic-gauge map of (3.3), and define

$$\tilde{F}_G = L \circ F_G, \quad L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right).$$

Then

$$\det D\tilde{F}_G = 1, \quad \operatorname{gdeg}(\tilde{F}_G) = N, \quad \max_i \deg(\tilde{F}_G)_i \leq 6N + 2.$$

The Galois closure of the induced degree- N function-field extension has geometric and arithmetic Galois group S_N . At

$$y_{P,a} = \left(1, 0, -\frac{2P(a)}{P'(a)} \right) \tag{2.1}$$

its full fiber is

$$\boxed{\tilde{F}_G^{-1}(y_{P,a}) \simeq \operatorname{Spec} K[T]/(P).}$$

Moreover, $y_{P,a}$ lies outside the nonproperness hypersurface of $\tilde{F}_G$, hence in its maximal finite-étale locus. More explicitly, for every commutative K -algebra R there is a bijection

$$\operatorname{Hom}_{K\text{-alg}}(K[T]/(P), R) \simeq \{u \in R^3 : \tilde{F}_G(u) = y_{P,a}\},$$

natural in R .

Consequently, every finite étale K -algebra of rank at least three occurs as a full fiber of a Jacobian-one map in affine three-space.

Proof. The polynomials P' and P''' are nonzero in characteristic zero when $N \geq 3$. Their product has only finitely many roots, while K is infinite, so an admissible a exists. The linear and cubic Taylor coefficients of G are $P'(a)$ and $P'''(a)/3!$, and the quadratic gauge of Section 3 is therefore defined.

At the target (2.1), its inverse polynomial is

$$G(S) - \frac{P'(a)}{2} \left(-\frac{2P(a)}{P'(a)} \right) = P(a + S).$$

Translation preserves squarefreeness, so Corollary 3.3 identifies the full fiber with $\operatorname{Spec} K[S]/(P(a + S))$.

The translation back to the original polynomial is explicit:

$$K[S]/(P(a + S)) \longrightarrow K[T]/(P(T)), \quad \overline{S} \longmapsto \overline{T} - a,$$

with inverse $\overline{T} \mapsto \overline{S} + a$. Composing this with the natural equivalence of Corollary 3.3 gives, for every commutative K -algebra R ,

$$\mathrm{Hom}_{K\text{-alg}}(K[T]/(P), R) \simeq \tilde{F}_G^{-1}(y_{P,a})(R).$$

Thus the isomorphism includes the complete scheme structure and commutes with base change.

The inverse polynomial at the target is $P(a + S)$, so its discriminant is nonzero. Corollary 3.10 therefore gives the asserted structural position in the maximal finite-étale locus.

The linear map L fixes the chosen target because its second coordinate is zero. It multiplies the Jacobian by $-1/2$ and changes neither geometric degree nor coordinate degree. Corollary 3.7 therefore gives the determinant-one statement and the bound $6N + 2$. Since both the quotient algebra and the generic inverse problem have degree N , the fiber is full. The symmetric-monodromy assertion follows from Theorem 3.4.

For a general finite étale algebra of rank at least three, use the monogenicity lemma in Section 4 and apply the polynomial case just proved. $\square$

Theorem 2.4 (Classification of full Keller fibers). *For $N \geq 1$, let $\mathrm{FullFib}_{3,N}(K)$ be the replete full subcategory of finite étale K -schemes whose objects are the schemes isomorphic to full fibers*

$$F^{-1}(y), \quad F : \mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3, \quad \mathrm{gdeg}(F) = N, \quad y \in \mathbb{A}^3(K).$$

Then

$$\mathrm{FullFib}_{3,N}(K) = \begin{cases} \mathrm{FinEt}_K^N, & N = 1 \text{ or } N \geq 3, \\ \emptyset, & N = 2, \end{cases}$$

where FinEt_K^N is the full subcategory of rank- N finite étale K -schemes. In particular, for $N \geq 3$, a finite K -scheme occurs as a full fiber of a geometric-degree- N Keller map in $\mathbb{A}_K^3$ if and only if it is finite étale of rank N .

Equivalently on coordinate algebras, every rank- N finite étale K -algebra occurs for $N = 1$ or $N \geq 3$, and none occurs for $N = 2$. Consequently, the possible nonzero ranks of full Keller fibers are exactly

$$\boxed{1, 3, 4, 5, \dots}.$$

Proof. The Jacobian condition makes F étale. Hence every fiber over a K -point is a finite étale K -scheme, and the fullness equality makes its rank equal to $\mathrm{gdeg}(F) = N$. This proves the necessary direction in every degree. The identity map realizes the unique rank-one object. Theorem 2.3 realizes every rank- N finite étale object in affine three-space for every $N \geq 3$, while Lemma 2.2 shows that no geometric-degree-two Keller map exists. $\square$

Remark 2.5 (What the categorical statement remembers). *The category $\mathrm{FullFib}_{3,N}(K)$ is the essential image inside finite étale K -schemes: it remembers the resulting fiber scheme, not a chosen presentation by a map–target pair (F, y) . Different pairs may realize the same object. If one works with finite étale algebras rather than schemes, the corresponding statement is contravariant under Spec .*

3 The quadratic gauge

The prescribed-algebra realization theorem rests on one explicit construction. Let

$$G(S) = g_1S + g_2S^2 + \cdots + g_NS^N \in K[S], \quad g_1g_3g_N \neq 0. \quad (3.1)$$

In source coordinates (x, y, z) define

$$t = 1 + xy, \quad q = t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t). \quad (3.2)$$

The associated map $F_G = (\Pi, B, C) : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ is

$$\begin{aligned} \Pi &= tq, \\ B &= y + 3\frac{g_3}{g_1} xq + 2\frac{g_2}{g_1} tq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k, \\ C &= x(5 - 3t) - \frac{g_3}{g_1} x^3 z - \sum_{k=4}^N (k-2) \frac{g_k}{g_1} (xq)^k. \end{aligned} \quad (3.3)$$

Although the formula is long, it is polynomial and completely mechanical in the coefficients of G .

The smallest seed recovers the counterexample that motivated this work. Take

$$G(S) = S + S^3.$$

Then

$$q = (1 + xy)^2 z + y^2 (4 + 3xy),$$

and (3.3) becomes, coordinate for coordinate,

$$\begin{aligned} \Pi &= (1 + xy)^3 z + y^2 (1 + xy) (4 + 3xy), \\ B &= y + 3x(1 + xy)^2 z + 3xy^2 (4 + 3xy), \\ C &= 2x - 3x^2 y - x^3 z. \end{aligned} \quad (3.4)$$

This is exactly the map announced by Alpöge [1]: there is no change of source coordinates, target coordinates, or scaling hidden in the comparison. In particular, the general determinant theorem below specializes directly to its Jacobian determinant -2 .

The point of the construction appears in the inverse problem. For a target (Π, B, C) put

$$G_\Pi(S) = g_1 S + \Pi(g_2 S^2 + g_3 S^3) + \sum_{k=4}^N g_k \Pi^k S^k \quad (3.5)$$

and

$$E_{\Pi, B, C}(S) = G_\Pi(S) - \frac{g_1}{2} (BS^2 + C). \quad (3.6)$$

Thus a three-variable inverse problem reduces to the single equation $E_{\Pi, B, C}(S) = 0$.

For $(\pi, b, c) \in K^* \times K^2$, write $\mathcal{F}_{\pi, b, c}$ for the fiber functor

$$\mathcal{F}_{\pi, b, c}(A) = \{(x, y, z) \in A^3 : F_G(x, y, z) = (\pi, b, c)\}$$

on commutative K -algebras.

Lemma 3.1 (Derivative units over test algebras). *Let $E \in K[S]$ be separable, let A be a commutative K -algebra, and let $s \in A$ satisfy $E(s) = 0$. Then $E'(s)$ is a unit of A .*

Proof. Choose $U, V \in K[S]$ with $UE + VE' = 1$. Evaluating at s gives $V(s)E'(s) = 1$. Since A is commutative, this is a two-sided inverse. $\square$

Theorem 3.2 (General localized quadratic-gauge fiber theorem). *The map F_G satisfies*

$$\det DF_G = -2, \quad \text{gdeg}(F_G) = N.$$

Its generic inverse equation is $E_{\Pi,B,C}(S) = 0$. Fix $(\pi, b, c) \in K^ \times K^2$, put $E = E_{\pi,b,c}$, and let $\overline{E'}$ be the class of E' in $K[S]/(E)$. Then*

$$\boxed{F_G^{-1}(\pi, b, c) \simeq \text{Spec}((K[S]/(E))[1/\overline{E'}])}. \quad (3.7)$$

Equivalently, for every commutative K -algebra A there is a natural bijection

$$\text{Hom}_{K\text{-alg}}((K[S]/(E))[1/\overline{E'}], A) \simeq \mathcal{F}_{\pi,b,c}(A). \quad (3.8)$$

Proof. We begin on the open set $t \neq 0$. Introduce the coordinates

$$S = \frac{x}{t}, \quad Q = y + xq, \quad D = 1 - SQ + \Pi S^2 = t^{-1}.$$

A direct differentiation gives

$$\det \frac{\partial(\Pi, S, Q)}{\partial(x, y, z)} = D^{-1}. \quad (3.9)$$

Next define

$$\beta(\Pi, S) = \frac{G'_\Pi(S)/g_1 - 1 - \Pi S^2}{S}. \quad (3.10)$$

The numerator is divisible by S . Substituting the definitions into (3.3) gives the two identities

$$B = Q + \beta(\Pi, S), \quad C = \frac{2G_\Pi(S)}{g_1} - BS^2. \quad (3.11)$$

The second identity is precisely $E_{\Pi,B,C}(S) = 0$. Differentiating it gives

$$D = \frac{G'_\Pi(S)}{g_1} - BS = \frac{E'_{\Pi,B,C}(S)}{g_1}. \quad (3.12)$$

At fixed Π , the Jacobian of (3.11) with respect to (S, Q) is

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2D. \quad (3.13)$$

The factors D^{-1} and $-2D$ cancel. Hence $\det DF_G = -2$ on the dense open set $t \neq 0$, and therefore everywhere.

We now identify the entire fiber. Let A be an arbitrary commutative K -algebra. An A -point (x, y, z) above (π, b, c) satisfies

$$tq = \pi.$$

Because π is a unit, both t and q are units. The chart coordinates

$$S = xt^{-1}, \quad Q = y + xq, \quad D = t^{-1}$$

are therefore defined on every point of the fiber. Equations (3.11) and (3.12) give

$$Q + \beta(\pi, S) = b, \quad E(S) = 0, \quad D = \frac{E'(S)}{g_1}.$$

Thus every source point gives a root S of E for which $E'(S)$ is a unit.

Conversely, suppose $s \in A$ satisfies $E(s) = 0$ and $E'(s) \in A^*$. Put

$$d = \frac{E'(s)}{g_1},$$

and define

$$Q = b - \beta(\pi, s), \quad t = d^{-1}, \quad x = sd^{-1}, \quad y = Q - \pi s,$$

followed by

$$q = \pi d, \quad z = d^2 \left(q - \frac{g_1}{g_3} y^2 (1 + 3t) \right).$$

The derivative identity says exactly that

$$d = 1 - sQ + \pi s^2.$$

A calculation in the ring A now gives

$$1 + xy = t, \quad t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t) = q, \quad tq = \pi.$$

The first identity in (3.11) gives $B = b$, and the root equation gives $C = c$. We have reconstructed a point of the original fiber.

The two procedures undo one another. Starting with a source point, one recovers

$$D = t^{-1}, \quad x = S/D, \quad y = Q - \pi S, \quad q = \pi D,$$

and then z from the definition of q . Starting with a root, the marked coordinate is again s . All formulas use ring operations and the inverse of the explicitly specified unit $E'(s)$, so they commute with homomorphisms of test algebras.

The universal property of $K[S]/(E)$ identifies its maps to A with roots of E in A . Localizing at $\overline{E'}$ imposes exactly the additional condition that $E'(s)$ be a unit. This proves (3.8), and Yoneda's lemma gives (3.7). Notice that this general localized statement requires no separability hypothesis: it removes precisely the ramified part of the inverse equation.

It remains to check the generic degree. Put

$$L = K(\Pi, B), \quad \lambda = \frac{g_1}{2},$$

and write

$$E_{\Pi, B, C}(S) = H_{\Pi, B}(S) - \lambda C \in L[C][S].$$

This polynomial has degree N in S . Its leading coefficient is $g_3 \Pi$ when $N = 3$ and $g_N \Pi^N$ when $N \geq 4$, so it is a unit of $L[C]$. Hence $E_{\Pi, B, C}$ is primitive over the UFD $L[C]$.

Assume it factors nontrivially over $L(C)[S]$. Gauss's lemma then gives a factorization in $L[C][S]$. Since the polynomial is linear in C , one factor, say $A(S)$, is independent of C . Comparing the coefficient of C shows that $A(S)$ divides the nonzero scalar $-\lambda$ in $L[S]$. Thus A is a unit, a contradiction. Therefore $E_{\Pi, B, C}$ is irreducible of degree N over $K(\Pi, B, C)$. The generic reconstruction identifies the source function field with the corresponding degree- N extension, and hence $\text{gdeg}(F_G) = N$. $\square$

Corollary 3.3 (Finite-étale (squarefree) quadratic-gauge fibers). *In the setting of Theorem 3.2, suppose that $E = E_{\pi,b,c}$ is squarefree of degree N . Then the localization disappears and, for every commutative K -algebra A , there is a natural bijection*

$$\mathrm{Hom}_{K\text{-alg}}(K[S]/(E), A) \simeq \mathcal{F}_{\pi,b,c}(A). \quad (3.14)$$

Consequently,

$$F_G^{-1}(\pi, b, c) \simeq \mathrm{Spec} K[S]/(E),$$

and this is a full finite-étale Keller fiber of rank N .

Proof. Since E is squarefree, choose $U, V \in K[S]$ with $UE + VE' = 1$. Modulo E , the class of V is an inverse for $\overline{E'}$, so the localization in Theorem 3.2 is just $K[S]/(E)$. The natural equivalence and scheme isomorphism follow from that theorem. Its geometric-degree statement makes the rank- N fiber full. $\square$

Theorem 3.4 (Full symmetric generic monodromy). *For every seed (3.1), not merely for a generic choice of its coefficients, the geometric and arithmetic monodromy groups of F_G are*

$$\boxed{\mathrm{Mon}_{\mathrm{geom}}(F_G) = \mathrm{Mon}_{\mathrm{arith}}(F_G) = S_N.}$$

Equivalently, the splitting field of $E_{\Pi,B,C}(S)$ over $K(\Pi, B, C)$ has Galois group S_N , and the same is true after replacing K by an algebraic closure. The conclusion is unchanged for the Jacobian-one normalization $\tilde{F}_G$.

Proof. We work first over $\overline{K}$. Fix $p \in \overline{K}^*$ and put $H(S) = G_p(S)$. Consider

$$f_b(S) = H(S) - \frac{g_1 b}{2} S^2.$$

We claim that f_b is a Morse polynomial for all but finitely many $b \in \overline{K}$: its $N - 1$ critical points are then simple and have pairwise distinct critical values.

A repeated root r of

$$H(S) - \frac{g_1}{2}(BS^2 + C)$$

cannot have $r = 0$, because $H'(0) = g_1$. The repeated-root locus is parametrized by

$$B(r) = \frac{H'(r)}{g_1 r}, \quad C(r) = \frac{2H(r) - rH'(r)}{g_1}, \quad r \in \mathbb{G}_m. \quad (3.15)$$

Differentiating gives

$$\frac{dC}{dB} = -r^2. \quad (3.16)$$

Let $M = \overline{K}(B(r), C(r)) \subset \overline{K}(r)$. Equation (3.16) shows that $r^2 \in M$. Since $[\overline{K}(r) : \overline{K}(r^2)] = 2$, either $M = \overline{K}(r^2)$ or $M = \overline{K}(r)$. The first possibility is excluded because $B(r)$ has a pole of odd order one at $r = 0$, whereas every element of $\overline{K}(r^2)$ has even order there. Thus $M = \overline{K}(r)$, so (3.15) is birational onto the reduced discriminant curve.

The rational map $r \mapsto B(r)$ has degree $N - 1$: its only poles have orders one at 0 and $N - 2$ at infinity. Away from finitely many values of b , its fiber therefore consists of $N - 1$ distinct points, which are precisely the critical points of f_b . At such a point,

$$f_b(r) = H(r) - \frac{rH'(r)}{2} = \frac{g_1}{2} C(r).$$

The birationality just proved implies that, after excluding finitely many more values of b , these critical values are pairwise distinct. This proves the claim.

Choose one such b . The classical Morse-polynomial theorem [11, Theorem 4.4.5] gives

$$\mathrm{Gal}(f_b(S) - T / \overline{K}(T)) = S_N.$$

Indeed, the polynomial cover is connected, its finite branch inertia groups are transpositions, and those inertia groups generate; a transitive permutation group generated by transpositions is the full symmetric group. After the substitution $T = g_1 C/2$, this cover is the restriction of the generic inverse cover to the line $\Pi = p$, $B = b$. Its monodromy is therefore a subgroup of the generic geometric monodromy. Hence

$$S_N \leq \mathrm{Mon}_{\mathrm{geom}}(F_G) \leq S_N,$$

which proves equality. Finally,

$$\mathrm{Mon}_{\mathrm{geom}}(F_G) \triangleleft \mathrm{Mon}_{\mathrm{arith}}(F_G) \leq S_N$$

forces the arithmetic group to be S_N as well. A linear automorphism of the target does not change either monodromy group, so the same conclusion holds for $\tilde{F}_G$. $\square$

Corollary 3.5 (Hilbertian specializations). *Suppose that K is Hilbertian. For every seed (3.1), there are infinitely many targets in $K^* \times K^2$ whose fibers are connected full finite-étale schemes of rank N and whose splitting fields have Galois group S_N over K .*

Proof. Choose $p \in K^*$. The proof of Theorem 3.4 excludes only finitely many values of b in an algebraic closure, so some $b \in K$ makes f_b Morse. Its splitting field over $K(C)$ is a regular S_N -extension by [11, Theorem 4.4.5]. Hilbert irreducibility [11, Chapter 3] supplies infinitely many $c \in K$ for which $E_{p,b,c}$ remains irreducible and has Galois group S_N . Such a polynomial is squarefree of degree N , so Corollary 3.3 identifies it with a connected full finite-étale fiber. $\square$

Remark 3.6 (Which hypotheses are used where). *The general localized quadratic-gauge fiber theorem, Theorem 3.2, uses only that π , g_1 , and g_3 are nonzero for the fiber identification, together with the identity*

$$E'(S) = g_1(1 - bS + S\beta(\pi, S) + \pi S^2).$$

Squarefreeness is needed only to conclude that E' is invertible at every root and hence that no localization remains. The condition $g_3 \neq 0$ is what makes the source coefficient g_1/g_3 meaningful. For a translated polynomial $P(a + S) - P(a)$, the cubic coefficient is the third Hasse derivative at a , equal to $P'''(a)/3!$ in characteristic zero. The Lean development follows the same separation.

The transposition inertia in Theorem 3.4 does not contradict the fact that F_G is étale. A repeated inverse root has $E' = 0$, so the reconstruction theorem removes it from the affine source; the corresponding valuation is centered on the nonproperness boundary. Monodromy is taken on the function-field cover, or equivalently on the finite étale restriction over a sufficiently small target open set.

Corollary 3.7 (Jacobian-one effective form). *Let*

$$L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right), \quad \tilde{F}_G = L \circ F_G.$$

Then

$$\det D\tilde{F}_G = 1, \quad \mathrm{gdeg}(\tilde{F}_G) = N.$$

For every $(\pi, b, c) \in K^* \times K^2$,

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \text{Spec} \left((K[S]/(E_{\pi, b, c})) [1/\overline{E'_{\pi, b, c}}] \right).$$

If $E_{\pi, b, c}$ is squarefree of degree N , this reduces to

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \text{Spec } K[S]/(E_{\pi, b, c}).$$

In particular, a target with $b = 0$ is unchanged by the normalization. Moreover,

$$\deg \Pi \leq 7, \quad \deg B \leq 6N + 2, \quad \deg C \leq 6N,$$

so

$$\boxed{\max_i \deg(\tilde{F}_G)_i \leq 6N + 2.}$$

Proof. The linear map L has determinant $-1/2$, so the determinant, geometric degree, and localized-fiber statements follow from Theorem 3.2; the squarefree simplification follows from Corollary 3.3.

For the degree estimate, $\deg t = 2$ and $\deg q \leq 5$, hence $\deg \Pi \leq 7$. A summand of B indexed by $k \geq 4$ has degree at most

$$2 \deg t + (k - 2) + k \deg q \leq 4 + (k - 2) + 5k = 6k + 2.$$

The preceding terms have degrees at most 1, 6, and 7, so $\deg B \leq 6N + 2$. Similarly, $(xq)^k$ has degree at most $6k$, while the remaining terms of C have degree at most 4. Thus $\deg C \leq 6N$. A linear output normalization does not change these bounds. $\square$

3.1 Checking the formula coefficient by coefficient

The compact formulas in the general localized quadratic-gauge fiber theorem, Theorem 3.2, hide a simple term-by-term calculation. We include it here because this is where the map for an arbitrary degree- N polynomial is assembled. The argument treats one generic coefficient g_k at a time; it is not inferred from low-degree examples.

From (3.5) and (3.10),

$$\beta(\Pi, S) = 2 \frac{g_2}{g_1} \Pi + \left(3 \frac{g_3}{g_1} - 1 \right) \Pi S + \sum_{k=4}^N k \frac{g_k}{g_1} \Pi^k S^{k-2}. \quad (3.17)$$

On the source chart,

$$S = \frac{x}{t}, \quad \Pi = tq, \quad Q = y + xq,$$

and therefore, for every $k \geq 4$,

$$\Pi^k S^{k-2} = t^2 x^{k-2} q^k. \quad (3.18)$$

Substitution now gives

$$\begin{aligned} Q + \beta(\Pi, S) &= y + xq + 2 \frac{g_2}{g_1} tq + \left(3 \frac{g_3}{g_1} - 1 \right) xq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k \\ &= y + 3 \frac{g_3}{g_1} xq + 2 \frac{g_2}{g_1} tq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k, \end{aligned}$$

which is the displayed coordinate B .

For the third coordinate, write

$$C_* = \frac{2G_\Pi(S)}{g_1} - BS^2.$$

Using $B = Q + \beta(\Pi, S)$ and the definition of β , we obtain

$$C_* = 2S - QS^2 + \left(1 - \frac{g_3}{g_1}\right) \Pi S^3 - \sum_{k=4}^N (k-2) \frac{g_k}{g_1} \Pi^k S^k. \quad (3.19)$$

The terms with $k \geq 4$ already have the desired source form, since

$$\Pi^k S^k = (xq)^k. \quad (3.20)$$

It remains to simplify the cubic part. Put $r = g_3/g_1$. Then

$$\begin{aligned} 2S - QS^2 + (1-r)\Pi S^3 &= \frac{2xt - x^2y - rx^3q}{t^2} \\ &= \frac{x(t+1) - rx^3q}{t^2}, \end{aligned}$$

where the second line uses $xy = t - 1$. Since

$$q = t^2z + \frac{1}{r}y^2(1+3t),$$

this becomes

$$-rx^3z + \frac{x(t+1 - (t-1)^2(1+3t))}{t^2}.$$

The elementary identity

$$t+1 - (t-1)^2(1+3t) = t^2(5-3t) \quad (3.21)$$

therefore gives

$$2S - QS^2 + (1-r)\Pi S^3 = x(5-3t) - \frac{g_3}{g_1}x^3z.$$

Together with (3.19) and (3.20), this is exactly the displayed coordinate C .

We derived the identities after inverting t , but both sides are polynomials in $K[x, y, z]$. Localization at the nonzero polynomial t is injective, so the identities hold everywhere. At fixed Π , their differential has determinant

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2 \left(\frac{G'_\Pi(S)}{g_1} - BS \right) = -2D.$$

The source-chart determinant is D^{-1} , and the chain rule gives $\det DF_G = -2$. This is the complete cancellation: each coefficient g_k enters independently, so the same calculation works in every degree.

3.2 The exact nonproperness hypersurface

We now make the fullness condition intrinsic. Work geometrically over $\overline{K}$, and write $S_{F_G} \subset \mathbb{A}_{\overline{K}}^3$ for the nonproperness, or Jelonek, set of F_G . Thus $y \in S_{F_G}$ precisely when there is a sequence, equivalently an algebraic arc, escaping to infinity in the source whose image tends to y . Put

$$\Delta_G(\Pi, B, C) = \text{Disc}_S(E_{\Pi, B, C}(S)). \quad (3.22)$$

The discriminant is taken with $E_{\Pi, B, C}$ regarded as a degree- N polynomial, including on the slice where its leading coefficient vanishes.

Theorem 3.8 (Exact quadratic-gauge nonproperness). *For every seed (3.1),*

$$\boxed{S_{F_G} = V(\Delta_G)}. \quad (3.23)$$

More precisely, the following statements hold.

1. If $\pi \neq 0$ and the multiplicity partition of $E_{\pi, b, c}$ is $(m_1, \dots, m_s)$, then

$$\#F_G^{-1}(\pi, b, c) = \#\{i : m_i = 1\}, \quad N - \#F_G^{-1}(\pi, b, c) = \sum_{m_i \geq 2} m_i. \quad (3.24)$$

2. On $\Pi = 0$, the complete geometric fiber table is

<i>target stratum</i>	$\#F_G^{-1}(0, B, C)$	<i>lost sheets</i>
$B \neq 0, BC \neq 1$	3	$N - 3$
$B \neq 0, BC = 1$	1	$N - 1$
$B = 0$	1	$N - 1.$

(3.25)

3. For $N = 3$, the raw resultant has an extraneous leading-coefficient factor:

$$\text{Res}_S(E, E') = \text{unit} \cdot \Pi \Delta_G, \quad \Delta_G(0, B, C) = \frac{g_1^4}{4} B^2 (1 - BC).$$

Hence the entire plane $\Pi = 0$ is not a nonproperness component in degree three.

4. For $N \geq 4$, put

$$e_N = N^2 - 3N - 2.$$

Then

$$\Delta_G = \Pi^{e_N} D_G, \quad D_G(0, B, C) = \kappa_G B^2 (1 - BC), \quad \kappa_G \in K^*. \quad (3.26)$$

In particular,

$$S_{F_G} = V(\Pi) \cup V(D_G),$$

and the resultant and discriminant have the same reduced zero set.

Proof. Because F_G is étale, every geometric fiber is finite and reduced. Its generic cardinality is N . The standard properness criterion for a generically finite polynomial map says that it is proper at a target exactly when the sum of the local multiplicities in that fiber is the generic degree [5, Proposition 6]. Here all local multiplicities are one, so

$$y \notin S_{F_G} \iff \#F_G^{-1}(y) = N. \quad (3.27)$$

Suppose first that $\pi \neq 0$. Theorem 3.2 identifies the fiber with

$$\text{Spec}((\overline{K}[S]/(E_{\pi,b,c}))[1/\overline{E'_{\pi,b,c}}]).$$

Thus a distinct root contributes one affine point exactly when it is simple. This proves (3.24); it also shows that the fiber has cardinality N exactly when $\Delta_G(\pi, b, c) \neq 0$.

It remains to analyze $\Pi = 0$, where the inverse chart used above is not available. In the source, $\Pi = tq$. On $q = 0$, one has

$$B = y, \quad C = 2S - BS^2, \quad S = \frac{x}{t}.$$

The residual inverse equation is therefore

$$BS^2 - 2S + C = 0. \tag{3.28}$$

A root reconstructs on $q = 0$ precisely when $1 - BS \neq 0$. Hence this divisor contributes two points when $B \neq 0$ and $BC \neq 1$, no point when $B \neq 0$ and $BC = 1$, and one point when $B = 0$.

On $t = 0$, the relation $1 + xy = 0$ gives $y = -1/x$, while

$$q = \frac{g_1}{g_3}y^2, \quad B = -2y = \frac{2}{x}.$$

Thus $t = 0$ contributes one point when $B \neq 0$ and none when $B = 0$. Once x and y are fixed, the equation for C is linear in z with nonzero coefficient, so the point is unique. The divisors $t = 0$ and $q = 0$ do not meet. This proves (3.25).

For $N = 3$, specializing the degree-three discriminant at $\Pi = 0$ gives

$$\Delta_G(0, B, C) = \frac{g_1^4}{4}B^2(1 - BC).$$

Its zero set is exactly the two short-fiber strata in (3.25). Since the leading coefficient is $g_3\Pi$, the resultant has one additional factor Π .

Now let $N \geq 4$ and put $r = N - 3$. Over the generic DVR of the divisor $\Pi = 0$, the lower Newton polygon of E has vertices

$$(0, 0), \quad (2, 0), \quad (3, 1), \quad (N, N).$$

Consequently its root valuations, with multiplicities, are

$$0 \ (2), \quad -1 \ (1), \quad -\frac{r+2}{r} \ (r).$$

The first two blocks are exactly the affine $q = 0$ and $t = 0$ branches found above; the final block consists of the $N - 3$ missing sheets.

Using

$$\text{Disc}(E) = \text{lc}(E)^{2N-2} \prod_{i < j} (s_i - s_j)^2$$

and the three displayed root valuations gives

$$\text{ord}_{\Pi}(\Delta_G) = 2N(N-1) - 4 - 4(r+2) - 2(r+2) - (r-1)(r+2) = N^2 - 3N - 2.$$

The endpoint residual polynomials are separable because $g_1g_3g_N \neq 0$, so this order is exact. Their discriminants also give

$$\Pi^{-e_N} \Delta_G|_{\Pi=0} = \kappa_G B^2(1 - BC)$$

with $\kappa_G \neq 0$. For example, up to the sign convention for the discriminant,

$$\kappa_G = \pm \frac{r^r}{4} g_1^4 g_3^{r+1} g_N^{r-1}.$$

The fiber table and (3.27) now show that the zero set of Δ_G is exactly the nonproperness set. $\square$

Proposition 3.9 (Boundary normalization and explicit escape arcs). *On $\Pi \neq 0$, the reduced discriminant divisor is birationally parametrized by*

$$B = \frac{G'_\Pi(s)}{g_1 s}, \quad C = \frac{2G_\Pi(s) - sG'_\Pi(s)}{g_1}, \quad (\Pi, s) \in \mathbb{G}_m^2. \quad (3.29)$$

If s_0 is a root of multiplicity $m \geq 2$ over (π, b, c) , every one of the m lost sheets is represented, after a finite base change, by a Laurent-series source arc. With root displacement as uniformizer, its generic valuations are

$$\text{ord}(x) = -(m-1), \quad \text{ord}(q) = m-1, \quad \text{ord}(z) \geq m-1.$$

For the generic point of $\Pi = 0$, put $d = N-3$, $h = \gcd(d, 2)$. The d lost sheets form h boundary primes, each with ramification index d/h . After the common base change $\Pi = u^d$, their valuations are

$$\begin{array}{c|cccccccc} & S & D & t & x & y & q & z \\ \hline \text{ord}_u & -(N-1) & -(N+1) & N+1 & 2 & -2 & -4 & -(2N+6). \end{array} \quad (3.30)$$

On the special stratum $B = 0$, two further missing sheets have

$$\Pi = u^2, \quad S \sim au^{-1}, \quad a^2 = -\frac{g_1}{g_3}, \quad (\text{ord}_u x, \text{ord}_u y, \text{ord}_u q, \text{ord}_u z) = (-1, 1, 2, 2). \quad (3.31)$$

Together with the multiplicity-block arcs above, these account for every lost sheet on every stratum of (3.25).

Proof. Equations (3.29) are $E = E' = 0$ solved for B, C . Their birationality was proved in the monodromy argument around (3.15).

For a repeated root, set $s(u) = s_0 + u$ and keep π, b fixed while putting

$$c(u) = \frac{2G_\pi(s(u))}{g_1} - b s(u)^2, \quad d(u) = \frac{G'_\pi(s(u))}{g_1} - b s(u).$$

Then $E_{\pi, b, c(u)}(s(u)) = 0$ identically and $\text{ord}_u d = m-1$. The exact reconstruction formulas

$$x = \frac{s}{d}, \quad y = b - \beta(\pi, s) - \pi s, \quad q = \pi d, \quad z = \pi d^3 - \frac{g_1}{g_3} y^2 (d^2 + 3d)$$

give the stated valuations. Applying Newton–Puiseux to the other roots in the same multiplicity block gives all m arcs after a finite base change.

For $\Pi = 0$, the final Newton edge has horizontal length d and endpoint equation

$$g_3 + g_N Z^d = 0.$$

Equivalently its binomial model is

$$W^d = \text{unit} \cdot \Pi^{d+2}.$$

It has $h = \gcd(d, 2)$ primes, each of ramification index d/h . Normalizing by

$$v(\Pi) = \frac{d}{h}, \quad v(S) = -\frac{d+2}{h}$$

gives

$$v(D) = -\frac{d+4}{h}, \quad v(x) = \frac{2}{h}, \quad v(y) = -\frac{2}{h}, \quad v(q) = -\frac{4}{h}, \quad v(z) = -\frac{2d+12}{h}.$$

Multiplication by h , corresponding to the common base change $\Pi = u^d$, yields (3.30).

If $B = 0$, the first Newton edge changes from $(0, 0) - (2, 0)$ to $(1, 0) - (3, 1)$. Its two nonzero roots satisfy

$$\Pi = u^2, \quad S \sim au^{-1}, \quad g_1 + g_3 a^2 = 0.$$

Here D and $t = D^{-1}$ have order zero. Substitution in the exact reconstruction formulas gives (3.31). When $BC = 1$ with $B \neq 0$, the two additional missing sheets instead form the finite double-root block already covered by the first part of the proposition. $\square$

Corollary 3.10 (Intrinsic fullness of the distinguished fiber). *For the seed $G(S) = P(a + S) - P(a)$ used in Theorem 2.3, the distinguished target $y_{P,a}$ lies in the maximal open subset on which $\tilde{F}_G$ is finite étale of degree N .*

Proof. At $y_{P,a}$, the inverse polynomial is $P(a + S)$. Hence

$$\Delta_G(y_{P,a}) = \text{Disc}(P(a + S)) = \text{Disc}(P) \neq 0.$$

Theorem 3.8 places the target outside the nonproperness hypersurface. Proper and étale is finite étale, and the degree is N . $\square$

Remark 3.11 (Polynomial curves). *The parametrization (3.29), with s fixed, gives polynomial curves in Π covering the discriminant component, while the component $\Pi = 0$ is covered by affine lines. Thus the explicit geometry realizes directly the general theorem that a complex Jelonek set is a uniruled hypersurface covered by polynomial curves of bounded degree [5, 6].*

4 From a polynomial to an arbitrary finite étale algebra

The main theorem was stated for a squarefree polynomial. To reach an arbitrary finite étale algebra, we only need the following standard form of the primitive-element argument. It works for products of fields as well as for a single field.

Lemma 4.1 (Monogenicity). *Every finite étale algebra over an infinite field is generated by one element. Equivalently, if A has rank N , then*

$$A \simeq K[T]/(P)$$

for a squarefree polynomial $P \in K[T]$ of degree N .

Proof. For $\theta \in A$, let $\chi_\theta(T)$ be the characteristic polynomial of multiplication by θ on the K -vector space A , and set

$$\Delta(\theta) = \text{disc}(\chi_\theta).$$

This is a polynomial function on the affine space underlying A . After extending scalars to a separable closure, identify $A \otimes_K K^{\text{sep}}$ with $(K^{\text{sep}})^N$. In these coordinates,

$$\Delta(\theta_1, \dots, \theta_N) = \prod_{i < j} (\theta_i - \theta_j)^2,$$

so Δ is not the zero polynomial. Since K is infinite, some $\theta \in A$ satisfies $\Delta(\theta) \neq 0$. Its characteristic polynomial has N distinct geometric roots, and therefore its minimal polynomial has degree N . Hence $K[\theta] = A$, with squarefree minimal polynomial P . $\square$

Proposition 4.2 (Compatibility with extension of scalars). *Let $K \hookrightarrow K'$ be an extension of characteristic-zero fields. For a squarefree $P \in K[T]$ and an admissible translation parameter $a \in K$, the map $\tilde{F}_G$ of Theorem 2.3, after coefficientwise extension to K' , is exactly the map obtained from the same P and a over K' . Its distinguished fiber satisfies*

$$\tilde{F}_{G,K'}^{-1}(y_{P,a}) \simeq \text{Spec}((K[T]/(P)) \otimes_K K') \simeq \text{Spec } K'[T]/(P).$$

Proof. Every formula in (3.2)–(3.3), the output normalization L , and the target (2.1) is obtained from the coefficients of P and the element a by field operations. The construction therefore commutes with scalar extension. The displayed fiber identity is the usual base-change isomorphism for a polynomial quotient, combined with the prescribed-algebra realization theorem, Theorem 2.3. $\square$

Corollary 4.3 (Number fields as connected Keller fibers). *Every finite separable field extension L/K of degree at least three occurs as a connected full fiber of an explicit Jacobian-one polynomial map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ of coordinate degree at most $6[L : K] + 2$.*

Proof. Choose a primitive element of L/K and apply Theorem 2.3 to its separable minimal polynomial. $\square$

Corollary 4.4 (Arithmetic data are preserved). *Let A be a finite étale algebra of rank at least three over a number field. The realization supplied by Theorem 2.3 preserves the algebra A itself. In particular it preserves:*

- *connectedness and field decomposition;*
- *real signature;*
- *splitting field and Galois action on geometric points;*
- *the local algebras $A \otimes_K K_v$ at every place;*
- *for any fixed integral model, factorization types away from its bad primes;*
- *the existence or nonexistence of rational and local points.*

Thus an arithmetic example built as a finite étale algebra of rank at least three can be used without alteration as a Keller fiber. This works because the fiber is $\text{Spec } A$ itself, not merely a set with the same number of points. Proposition 4.2 gives the corresponding compatibility with local fields and other scalar extensions.

5 The smallest Hasse-principle failure

For a number field K , call a Keller fiber A a *Hasse failure* if $\text{Spec } A$ has a point over every completion of K but no K -rational point. Before constructing an example, we show that rank five is the first place where this can happen.

Lemma 5.1 (Degree-four barrier). *A nonempty finite étale $\mathbb{Q}$ -scheme of degree at most four which has a point over every $\mathbb{Q}_p$ has a rational point.*

Proof. Let X be such a scheme and let L split it. At every unramified prime, $X(\mathbb{Q}_p) \neq \emptyset$ exactly when the Frobenius element fixes a point of the finite $\text{Gal}(L/\mathbb{Q})$ -set $X(L)$.

Suppose first that X is connected of degree greater than one. The Galois action is transitive and contains a derangement. Indeed, Burnside's lemma says that the average number of fixed points is one, whereas the identity fixes more than one point. Some group element therefore fixes none. Chebotarev produces an unramified prime with that Frobenius class, contradicting local solubility.

Now suppose that X is disconnected and has no rational component. In total degree at most four, the only remaining possibility is a disjoint union of two quadratic fields. If they are equal, choose an inert prime. If they are distinct, their compositum has Galois group $C_2 \times C_2$ and contains an element nontrivial in both quadratic quotients. Chebotarev again gives a prime inert in both components. Thus every everywhere locally soluble finite étale scheme of degree at most four contains $\text{Spec } \mathbb{Q}$. $\square$

We use the classical polynomial of Berend and Bilu,

$$P_5(T) = (T^3 - 19)(T^2 + T + 1). \quad (5.1)$$

It appeared in their study of polynomials with roots modulo every integer [2].

Theorem 5.2 (Optimal Hasse-failing Keller fiber). *There is an explicit Keller map $F : \mathbb{A}_{\mathbb{Q}}^3 \rightarrow \mathbb{A}_{\mathbb{Q}}^3$ of geometric degree five and a rational target y such that*

$$F^{-1}(y)(\mathbb{Q}) = \emptyset, \quad F^{-1}(y)(\mathbb{Q}_v) \neq \emptyset \quad \text{for every place } v.$$

No Hasse-failing Keller fiber has rank below five. Hence

$$\boxed{d_{\text{HP}} = 5}.$$

Proof. The polynomial (5.1) has no rational root. Let $p \neq 3$ be prime. If $p \equiv 1 \pmod{3}$, then $T^2 + T + 1$ has a simple root modulo p . If $p \equiv 2 \pmod{3}$, cubing is bijective on $\mathbb{F}_p^*$, so $T^3 - 19$ has a simple root modulo p . At $p = 3$, write $T = 1 + 3u$. After division by 9, the equation $T^3 = 19$ becomes

$$u + 3u^2 + 3u^3 = 2.$$

It has the simple solution $u \equiv 2 \pmod{3}$ and therefore lifts. The cubic factor has a real root. Thus (5.1) has a root over every completion of $\mathbb{Q}$.

The map below has fiber exactly $\text{Spec } \mathbb{Q}[T]/(P_5)$, so its local and global points are the same. Lemma 5.1 proves that no smaller rank can work. $\square$

For reference, here is the actual map. It comes directly from the normalized gauge, so its determinant does not require a separate guess or search. Put

$$G(T) = P_5(T) - P_5(0) = T^5 + T^4 + T^3 - 19T^2 - 19T.$$

Thus

$$(g_1, g_2, g_3, g_4, g_5) = (-19, -19, 1, 1, 1).$$

With $t = 1 + xy$ and

$$q = t^2 z - 19y^2(1 + 3t),$$

Theorem 3.2 gives the gauge map

$$\begin{aligned}\Pi &= tq, \\ B_0 &= y - \frac{3}{19}xq + 2tq - \frac{4}{19}t^2x^2q^4 - \frac{5}{19}t^2x^3q^5, \\ C_0 &= x(5 - 3t) + \frac{1}{19}x^3z + \frac{2}{19}(xq)^4 + \frac{3}{19}(xq)^5,\end{aligned}$$

with $\det D(\Pi, B_0, C_0) = -2$. Apply the target linear automorphism

$$L(\Pi, B_0, C_0) = (\Pi, 19B_0, 19C_0).$$

The resulting denominator-free map $F = L \circ F_G$ is

$$\begin{aligned}F_1 &= tq, \\ F_2 &= 19y - 3xq + 38tq - 4t^2x^2q^4 - 5t^2x^3q^5, \\ F_3 &= 19x(5 - 3t) + x^3z + 2(xq)^4 + 3(xq)^5.\end{aligned}\tag{5.2}$$

Consequently,

$$\boxed{\det DF = \det(L)(-2) = 19^2(-2) = -722.}$$

The normalized target $(1, 0, -2)$ is sent to

$$y = (1, 0, -38).$$

At that target the inverse equation is

$$G(T) - \frac{-19}{2}(-2) = G(T) - 19 = P_5(T).$$

The finite-étale (squarefree) fiber corollary, Corollary 3.3, therefore gives

$$\boxed{F^{-1}(1, 0, -38) \simeq \operatorname{Spec} \mathbb{Q}[T]/((T^3 - 19)(T^2 + T + 1)).}\tag{5.3}$$

So the smallest Hasse-failing Keller fiber is not only existential; its map, target, determinant, and quotient algebra are all explicit.

6 Infinitely many failures in one fixed map

The prescribed-algebra realization theorem changes the Keller map when P changes. For Hasse failures we can do better: one fixed quadratic-gauge map already contains infinitely many of them.

Put

$$G_\infty(S) = S^5 - \frac{3}{2}S^4 + \frac{3}{2}S^3 - \frac{5}{4}S^2 + \frac{9}{16}S$$

and let $F_\infty = F_{G_\infty}$ be the map of the general localized quadratic-gauge fiber theorem, Theorem 3.2. Thus $\det DF_\infty = -2$.

Theorem 6.1 (One map, infinitely many Hasse failures). *For every rational prime $\ell \equiv 1 \pmod{27}$, put*

$$y_\ell = \left(1, \frac{32\ell}{9}, \frac{8\ell+1}{3}\right).$$

Then

$$F_\infty^{-1}(y_\ell) \simeq \operatorname{Spec} \mathbb{Q}[X]/((X^3 - \ell)(X^2 + X + 1)).$$

Each fiber has points over every completion of $\mathbb{Q}$ but no rational point. The targets are pairwise distinct, and with the standard projective height,

$$\#\{y_\ell : H(y_\ell) \leq B\} \sim \frac{B}{576 \log B}.$$

Proof. Set $X = S - \frac{1}{2}$. The identity

$$(S - \frac{1}{2})^3(S^2 + \frac{3}{4}) = G_\infty(S) - \frac{3}{32}$$

turns the inverse equation at y_ℓ into

$$(X^3 - \ell)(X^2 + X + 1).$$

This polynomial is squarefree and has no rational root. The local argument is the same as in Theorem 5.2. At $p = 3$, the stronger congruence $\ell \equiv 1 \pmod{27}$ provides a cubic root by Hensel's lemma. At $p = \ell$, the quadratic factor splits because $\ell \equiv 1 \pmod{3}$. Every other prime is covered by the same residue-class dichotomy as before. Dirichlet's theorem supplies infinitely many primes in the progression $1 \pmod{27}$.

The primitive projective coordinates of y_ℓ are $[9 : 9 : 32\ell : 24\ell + 3]$, so $H(y_\ell) = 32\ell$. The prime number theorem in the progression $1 \pmod{27}$ gives the stated asymptotic. $\square$

7 Consequences, checks, and scope

The main theorem is a prescribed-fiber realization result. Given a finite étale algebra, it builds a Keller map whose full fiber is that algebra. The classification by rank is a useful corollary, but it is not the main point: two finite étale algebras of the same rank can have completely different field decompositions, Galois actions, signatures, and local behavior, and the construction distinguishes all of that data.

Several consequences are immediate. A squarefree polynomial of degree at least three becomes an exact inverse polynomial after one translation; an irreducible polynomial gives a connected fiber; and a degree- N input is realized in $\mathbb{A}^3$ with coordinate degree at most $6N + 2$. Moreover, every map produced by the construction has geometric and arithmetic generic monodromy S_N . Over a Hilbertian field, each such fixed map has infinitely many connected full fibers with splitting-field group S_N .

The fullness condition also has an intrinsic geometric interpretation for this family. Theorem 3.8 computes the exact nonproperness hypersurface, and Corollary 3.10 shows that every prescribed target lies in its complement. Thus fullness is not only a length equality checked at the selected target: it is equivalent there to membership in the maximal finite-étale locus. The partition formula (3.24) and the special table (3.25) describe exactly how fullness fails on every boundary stratum.

More importantly, the construction does not forget the arithmetic structure. If the original algebra has a particular real signature, splitting field, Galois action, or collection of local factorizations,

the fiber has the same data. The same applies to local and rational points. Thus an intersective polynomial with no rational root gives a Hasse-failing Keller fiber directly, without adding auxiliary factors or extra variables.

This is a transfer statement, not an inverse Galois theorem. It does not claim that arbitrary local specifications can always be globalized. It says that once a finite étale algebra with the desired arithmetic properties exists, the Keller construction preserves it exactly. Nor does the monodromy theorem claim that one fixed seed realizes every rank- N finite étale algebra: the prescribed algebra is used to choose the seed. Its precise strengthening is that every resulting cover, including one built for a highly special algebra, has maximal generic monodromy.

The realization is also far from unique. Other constructions in the accompanying repository place the same finite étale algebra inside weighted, cancellation, and quadratic-gauge Keller maps that need not be stably equivalent. That is a different question, and I leave it for a separate paper.

Relation to earlier work. To my knowledge, the following prescribed-algebra statement has not appeared before: every finite étale K -algebra of rank at least three occurs as a *full scheme fiber* of an explicit Keller map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$. The rank set $\{1, 3, 4, \dots\}$ follows from this theorem together with the known degree-two obstruction. I searched for earlier results up to 24 July 2026, but this priority statement is necessarily qualified: searches can miss unpublished work or a theorem stated in different language.

The closest constructions have a different scope. Gallagher gives explicit Keller maps in every geometric degree together with complete split rational fibers [4]; these realize K^N , rather than an arbitrary finite étale K -algebra. Miranda–Neto studies radicality and maximal-ideal decompositions of fibers that already come from Keller maps [8]. Lipton and Markakis connect rational images with Hilbert irreducibility [7], and Shaska studies graded Keller maps and arithmetic thinness [10]. None of these results states the prescribed-algebra realization theorem used here. The monodromy argument uses the classical theorem that a Morse polynomial has symmetric Galois group over the value-function field [11, Theorem 4.4.5]. The global boundary argument uses Jelonek’s fiber-multiplicity criterion and is consistent with his theorem that the nonproperness locus of a dominant complex polynomial self-map is a uniruled hypersurface [5]. Jelonek and Lasoń later bounded the degrees of polynomial curves covering such a locus [6]; in the present family the curves and missing source arcs are explicit.

Independent computer checks. The formulas are uniform in the degree, but expanding them by hand is not a pleasant way to find a sign error. I therefore checked the construction in three independent ways.

The SymPy script

```
scripts/verify_universal_quadratic_gauge.py
```

verifies the source-chart identity, the marked-line formulas, the Jacobian cancellation, every generic coefficient contribution, and the degree bound. These are symbolic identities with arbitrary coefficients, not numerical tests at a few sample points.

The Singular script

```
scripts/verify_universal_quadratic_gauge.sing
```

starts again over a rational function field in six independent coefficients. It expands the complete degree-six map, obtains determinant exactly -2 , and checks the degree profile $(7, 38, 36)$. This gives a separate CAS implementation of the same construction.

Finally,

`scripts/verify_finite_etale_keller_fibers.py`

builds translated seeds in degrees three, four, and five, expands the actual maps, reconstructs the prescribed quotient rings, and checks the explicit rank-five Hasse example and the fixed-map family. These bounded examples are regression tests. The proof for arbitrary degree is the coefficientwise argument in the general localized quadratic-gauge fiber theorem, Theorem 3.2, and Subsection 3.1.

The additional checker

`scripts/verify_quadratic_gauge_nonproperness.py`

independently verifies the degree-three and degree-four discriminant and resultant factors, the exact Π -adic discriminant order through degree ten, the uniform Newton-polygon formula through degree sixty-four, the $q = 0$ and $t = 0$ special-fiber charts, and all displayed boundary valuations. The pre-existing normalization checker

`scripts/verify_quadratic_weighted_stable_separation.py`

derives the same all-degree boundary ledger from the finite normalization.

Lean formalization. The Lean project lives in

`formal/finite-etale-keller.`

It contains no `sorry` and introduces no project-specific axioms. The formal development covers the constructive core of the paper. Given a squarefree polynomial P of degree at least three, it:

1. proves that an admissible translation exists and chooses one;
2. constructs the complete all-degree map as an actual $\text{MvPolynomial}(\text{Fin } 3, K)$ expression;
3. proves that its normalized Jacobian is one and that all coordinate degrees are at most $6 \deg P + 2$;
4. proves that the inverse polynomial at the selected target is $P(a + S)$; and
5. identifies the literal map fiber naturally with $\text{Hom}_{K\text{-alg}}(K[T]/(P), A)$ for every commutative test algebra A , while recording that the representing quotient has K -dimension $\deg P$ and is finite étale over K .

It also checks that the seed $G(S) = S + S^3$ gives the announced map (3.4) coordinate for coordinate. It proves that the three map coordinates are algebraically independent and constructs the induced injective pullback on rational function fields. Promoting Π and B to independent parameters, it also proves irreducibility and degree $\deg G$ of the full inverse equation over the iterated presentation $K(\Pi, B)(C)$, and proves that its root quotient has that finrank. For the optimal quintic example it goes further: the exact denominator-free map (5.2), at the exact target $(1, 0, -38)$, has a literal fiber naturally represented over every commutative rational test algebra by

$$\mathbb{Q}[T]/((T^3 - 19)(T^2 + T + 1)),$$

and Lean proves that this quotient is finite étale and has rank five. It also proves that the literal fiber has no rational point and constructs a real point and a three-adic point, the latter by an explicit Hensel argument at the approximate root -2 .

The final construction and degree certificate are exposed as

`automaticRealizationMap_certificate,`

the full independent-parameter inverse certificates are

`generalGaugeFullyGenericInversePolynomial_certificate` and
`generalGaugeFullyGenericInverseAdjoinRoot_finrank,`

and the natural fiber equivalence is

`automaticJacobianOneFiberRepresentingEquiv_natural.`

The corresponding concrete quintic certificate is

`integralFiberRepresentingEquiv_natural.`

Their axiom reports contain only Lean’s standard logical principles `propext`, `Classical.choice`, and `Quot.sound`.

Lean does not currently cover every theorem used in the final classification. Lean proves algebraic independence and an injective function-field pullback. Over the iterated independent-parameter field $K(\Pi, B)(C)$, it proves that the full inverse equation is irreducible of degree N and that its root quotient has $\text{finrank } N$. It does not yet construct the explicit comparison between this inverse-root extension and the actual source-over-target pullback-field extension, which is the remaining formal step needed to identify the result with $\text{gdeg}(F_G)$. Lean also proves that the complete map and target attached to a supplied translation parameter commute coefficientwise with extension of the ground field, and proves the tensor-product base change of the representing polynomial quotient. Exact nonproperness and boundary-sheet accounting, the symmetric-monodromy and Hilbertian-specialization arguments, monogenicity of arbitrary finite étale products, and the Campbell–Razar–Wright degree-two theorem remain ordinary mathematical proofs. For the quintic example, Lean now verifies the exact literal fiber, its rank, finite étaleness, the absence of rational points, and the existence of a real point and a three-adic point, but not yet the existence of roots over every remaining nonarchimedean completion. The full finite-field case split, Chebotarev, Dirichlet, and prime-counting arguments in the arithmetic applications are likewise outside the formal development. I list this boundary explicitly because a green Lean build should not be read as a claim that those external inputs have also been formalized.

Provenance and AI assistance. The recent three-dimensional counterexample was announced by Alpöge, who credited Akhil Mathew for suggesting the problem and Fable for producing the example [1]. Gallagher independently developed a weighted all-degree viewpoint and a public atlas of complete rational fibers [4]. The present proof uses a root-engineered quadratic gauge that contains the announced map as the seed $S + S^3$. Berend and Bilu supplied the classical intersective quintic, while Campbell, Razar, and Wright supplied the degree-two obstruction [3, 9, 12].

I used generative AI tools during exploration, symbolic-checker development, proof review, literature search, Lean formalization, and manuscript editing. They were useful as fast but fallible collaborators. I checked the stated mathematical claims, rewrote the exposition, and take responsibility for the paper.

References

- [1] L. Alpöge, *Public announcement of a counterexample to the Jacobian conjecture*, 20 July 2026, X post 2079028340955197566.
- [2] D. Berend and Y. Bilu, *Polynomials with roots modulo every integer*, Proc. Amer. Math. Soc. **124** (1996), 1663–1671, doi:10.1090/S0002-9939-96-03210-8.

- [3] L. A. Campbell, *A condition for a polynomial map to be invertible*, Math. Ann. **205** (1973), 243–248, doi:10.1007/BF01349234.
- [4] A. Gallagher, *Counterexample atlas: generic fiber degrees 3 through 100*, online exact certificate atlas, 2026, jacobianfun.org.
- [5] Z. Jelonek, *The set of points at which a polynomial map is not proper*, Ann. Polon. Math. **58** (1993), 259–266, doi:10.4064/ap-58-3-259-266.
- [6] Z. Jelonek and M. Lasoń, *Quantitative properties of the non-properness set of a polynomial map*, Manuscripta Math. **156** (2018), 383–397, arXiv:1411.5011.
- [7] R. J. Lipton and E. Markakis, *Some remarks on the Jacobian conjecture and connections with Hilbert’s irreducibility theorem*, arXiv:math/0507525, 2005.
- [8] C. B. Miranda–Neto, *An ideal-theoretic approach to Keller maps*, Proc. Edinburgh Math. Soc. **62** (2019), 1033–1044, doi:10.1017/S0013091519000099.
- [9] M. Razar, *Polynomial maps with constant Jacobian*, Israel J. Math. **32** (1979), 97–106, doi:10.1007/BF02764906.
- [10] T. Shaska, *Graded Keller maps and the Jacobian conjecture*, arXiv:2607.20210, 2026.
- [11] J.-P. Serre, *Topics in Galois Theory*, second ed., Research Notes in Mathematics 1, A K Peters, Wellesley, MA, 2007.
- [12] D. Wright, *On the Jacobian conjecture*, Illinois J. Math. **25** (1981), 423–440, doi:10.1215/ijm/1256047158.